

At a press of a button, the machine conjures up food. The food is designed to address any nutritional deficiencies of the person the food is meant for. The project is still in the design phase, and nobody knows what kind of form the final product will take. The undertaking is code named "Iron Man".

06 GRAVITY BATTERY

Electricity generation is a tricky process as there are few effective long term ways to store electricity once it is produced. One of the solutions to excess electricity production, is called pumped hydro. Water in dams is pumped up to higher altitudes when there is plenty of production. In times of high demand, the water is allowed to fall down, producing hydro electricity in the process. A company called Energy Vault has developed a mechanism that uses the same principle, but uses concrete blocks instead of water. This is known as the Gravity Battery. When excess electricity is produced in wind or solar farms, a system of cranes stack up the blocks of concrete in a vertical tower. When the demand is high, the blocks go through a controlled descent, and the kinetic energy of the fall is used to produce electricity. Tata Power has agreed to buy this form of electricity storage from Energy Vault.

07 HOUSEHOLD ROBOTS

This is another one of those things that should have been around already. Turns out, there are a bunch of problems when it comes to putting a commercial household robot, that can perform all kinds of sundry tasks. One of them is that robots are just not very good at visually processing dynamic environments very quickly. The cleaning robots for example takes some time to map out the floors and struggles to adapt if too many things change too quickly. The problem is exacerbated in three dimensions. Bipedal robots that can use the appliances designed for humans are just not very good at standing or walking, simple actions such as opening a door can send them toppling. DARPA is funding research to solve this problem.

08 NANORADIO

Nanoradios are devices made up of single carbon nanotubes, that are one ten-thousandth the width of a single strand of human hair. This single tube acts as a tuner, amplifier and demodulator, which are all separate components on conventional radios. In 2007, researchers at the University of California, Berkeley built such a nanoradio, and used it to receive signals from a conventional FM station. In the future, such nanoradios could also be used to transmit signals. When that is done, medical devices can be made, by introducing the nanoradios into the bloodstream. The medical applications include better delivery of drugs targeting only cancerous cells, and monitoring the insulin level in those suffering from diabetes.



Croc airless tires

09 NON PNEUMATIC TIRES

Also known as airless tires, there are already used for lawn mowers, heavy construction equipment, military vehicles and golf carts. They are also used for some wheelchairs and bicycles. The main reason to use airless tires is that they reduce the cost of replacement and maintenance, as they do not lose air pressure, cannot burst and do not go flat. Michelin had developed a version of the tire that combines the wheel and a tire into a single structure known as the tewel. Because of the lack of a bladder, the treads can incorporate features that evacuate water and prevent hydroplaning. Additionally, airless tires are better at braking, as the tires can be designed to have more lateral

stiffness, but less vertical stiffness for shock absorption. Unfortunately, the vehicles tended to shake a lot and become unstable in speeds in excess of 80 km/h. Michelin has no current plans of commercialising the product. Bridgestone has also developed a similar tire, but that is in the prototype phase. Crocodile makes airless tires, but they are mostly used in low speed construction equipment.

10 ONCOLYTIC VIRUS

Once of the problems with conventional cancer therapies is that they indiscriminately attack healthy cells as well. Researchers are trying to engineer viruses such as herpes simplex, poliovirus and measles to selectively attack cancer cells. Some viruses, such as the reoviruses are non pathogenic in humans, and naturally kill cancer cells. In some countries, including China and the US, some oncolytic viruses have been cleared for treatment of melanoma, or skin cancer. One of the biggest obstacles of using oncolytic viruses, is that the body's immune system kicks in and tries to eradicate the virus. Some researchers have attempted to suppress the immune system of the body to make the virus more effective, but by and large, the improved immune response is considered beneficial, as it helps the body fight the growth of tumours better.

11 OPTICAL RECTENNA

An optical rectenna is an antenna that converts light in the visible or infrared spectrum into electricity. While there are rectifying antennas for microwaves and radio waves, there are not cheap and efficient ones developed for the visible and infrared portions of the electromagnetic spectrum. Theoretically, the only real roadblock is optimising the antenna for the frequency. Optical rectennas have been demonstrated in labs, mostly using high energy lasers, to produce small amounts of electricity. Improvements in the technology has the potential to significantly boost the productivity of solar farms, by improving the efficiency of solar panels. The roadblocks have to do with the